NILESH BALU

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Education

- **Master of Science in Mechanical Engineering** *Expected Feb 2027*
  Eidgenössische Technische Hochschule (ETH) Zurich, Switzerland CGPA: 5.91/6.00 (ongoing)

- **Bachelor of Technology (Honors) in Mechanical Engineering; Minor in Systems Engineering** *July 2024*
  Indian Institute of Technology (IIT) Madras, India CGPA: 9.69/10.00
  – Gold medal for best academic record in B.Tech Mechanical Engineering
  – Gold medal for best individual project in B.Tech Mechanical Engineering

Academic Honors & Awards

- **IIT Madras Academic Prizes (2020 – 2024)**: Recipient of five distinct memorial awards and department prizes for maintaining exemplary academic performance throughout the B.Tech Mechanical Engineering program.
- **Mitacs Globalink Research Internship Award (2023)**: Selected for a 12-week internship at *Ontario Tech University*, Canada, to conduct research as part of a highly competitive global cohort.
- **JEE Advanced (2020)**: Secured an All India Rank of 1082 out of 150,000 candidates (Top 0.7% nationwide) in the Joint Entrance Examination for admission to the IITs.
- **KVPY Fellowship, SX Stream (2019)**: Awarded the *Kishore Vaigyanik Protsahan Yojana* Fellowship by the Government of India. Secured an All India Rank of 1492 out of 50,000 candidates (Top 3% nationwide).

Publications

1. Tashiro, N. K., Ballardini, G., **Balu, N.**, & Kuchenbecker, K. J. (2026, July). Human Perception of Asymmetric Vibrations under Diverse Grip Forces [Work-in-progress paper]. EuroHaptics 2026.

Research & Engineering Experience

- **Master's Thesis: Safe Flow Matching for Robot Trajectory Generation** *Jul 2026 - Present*
  Guided by Prof. Lars Lindemann ETH Zurich
  – Formulated a hard-constrained flow matching framework that optimizes auxiliary control inputs on the sampling dynamics via *Safe Gradient Flows*, enforcing constraints on the generated trajectory rather than on intermediate latents; each step reduces to a dual problem with a closed-form solution.
  – Proved anytime safety guarantees using control barrier function theory under standard constraint qualifications, every iterate satisfies the safety constraints.
  – Demonstrated zero executed constraint violations and faster planning than state-of-the-art baselines in preliminary MuJoCo experiments on robot navigation; Currently extending to further tasks and real-robot experiments.

- **Research Intern – Haptics and Dynamical Modeling** *Aug 2025 - Dec 2025*
  Guided by Prof. Katherine J. Kuchenbecker Max Planck Institute for Intelligent Systems
  – Investigated a haptic illusion in which asymmetric vibrations of a voice-coil actuator create a perceived pulling force: built a lumped-parameter model, identified its parameters from experimental data, designed feedforward controllers to track reference acceleration profiles, and designed a user study on how grip force affects the illusion (EuroHaptics 2026 work-in-progress paper).
  – Modeled and identified the dynamics of a novel fingertip haptic actuator from experimental data to simulate the high-frequency vibrations it induces.

- **Semester Thesis: Hand Pose Estimation for Digital Surgical Twins** *Feb 2025 - Jul 2025*
  Guided by Prof. Mirko Meboldt ETH Zurich
  – Developed a real-time hand pose estimation pipeline for surgical twins using stereo vision and deep learning; deployed WiLoR on Jetson Orin to extract 2.5D keypoints and triangulated them into 3D coordinates via numerical optimization. Enhanced accuracy and temporal consistency using Kalman filtering and outlier rejection, achieving sub-centimeter error.
  – Validated the pipeline against ArUco markers, infrared motion capture, and optical blob tracking, demonstrating efficacy for surgical training and motion analysis.

- **Bachelor's Thesis Project: Wearable Hand Assistive Device** *Aug 2023 - Aug 2024*
  Guided by Prof. Manish Anand IIT Madras
  – Developed a proof-of-concept assistive hand exoskeleton delivering up to 10 N grasping force for activities of daily living (ADL). Implemented an *underactuated tendon mechanism* to drive digital flexion, reducing mass and footprint while providing compliance.

– Implemented an EMG-based intent classification pipeline to automate flexion and extension cycles, providing two distinct grip modalities: *cylindrical grasp* and *lateral pinch*.

- **Research Intern – Robotics** *May 2023 - Aug 2023*
  Mechatronic and Robotic Systems Laboratory: Guided by Prof. Scott Nokleby Ontario Tech University, Canada
  – Engineered a proof-of-concept autonomous electric wheelchair for indoor navigation using ROS to integrate hardware interfaces and control loops. Implemented GMapping for SLAM with ZED cameras, Roboteq motor drivers, and Arduinos for real-time environment mapping and localization.
  – Enhanced odometry accuracy via sensor fusion by deploying an Unscented Kalman Filter (UKF) combining wheel encoders and IMU data; achieved autonomous navigation using Adaptive Monte Carlo Localization (AMCL) and RRT-based path planning.
- **Research Intern – Mechatronics and Sensor Integration** *Jul 2022 - May 2023*
  Rehabilitation Research and Device Development (R2D2) Laboratory: Guided by Prof. Manish Anand IIT Madras, India
  – Prototyped a 3-in-1 modular wheelchair, reconfigurable into scooter-board, indoor powered wheelchair, and outdoor handlebar-driven configurations.
  – Designed an ergonomic wrist module housing an ICM42688 IMU on a custom PCB for gesture- and tap-based speed control, alongside a wireless ESP32 throttle module for PID speed control of a BLDC motor.

Coursework Projects

- **On the Parallelization of Seam Carving** *Mar 2024 - May 2024*
  Course Project: Parallel Scientific Computing
  – Implemented a parallelized *Seam Carving* operator for content-aware image retargeting and object removal.
  – Benchmarked parallel scaling across *OpenMP, MPI, and OpenACC* frameworks and evaluated speedup and computational efficiency.
- **Simulation of a Four-Tank System using Model Predictive Control** *Oct 2022 - Nov 2022*
  Course Project: Modern Control Theory
  – Formulated, implemented, and tuned a *Model Predictive Controller (MPC)* in MATLAB for a benchmark four-tank system.
  – Derived state-space dynamics and implemented both *Kalman filter* and *Particle filter* observers, conducting a comparative analysis of state estimation robustness.

Teaching & Student Assistant (HiWi) Positions

- **RobotX Robotics Summer School 2026** *Jan 2026 - Jun 2026*
  Teaching Assistant ETH Zurich
  – Developed a high-fidelity MuJoCo simulation environment for the Unitree A2 quadruped, integrating a multi-modal sensor suite (IMU, joint proprioception, 3D LiDAR, and RGB-D cameras).
  – Trained robust PPO locomotion policies for flat- and rough-terrain traversal using Unitree-RL-MJLab.
- **Programming for Robotics – Introduction to ROS** *Jan 2026 - Feb 2026*
  Teaching Assistant ETH Zurich
  – Designed and implemented course assignments focused on ROS 2 navigation and perception, providing technical mentorship to students during exercise sessions.
- **RobotX Robotics Summer School 2025** *Jan 2025 - Jul 2025*
  Teaching Assistant ETH Zurich
  – Ported the legacy software stack from *ROS* to *ROS 2*, implementing a robust low-level controller interface and a real-time object detection/localization node for autonomous artifact identification.

Skills

- **Analytical Skills**: Laplace Analysis, Fourier Analysis, Principal Component Analysis, Linear Estimation Theory, Model Predictive Control, Finite Volume Method, Finite Element Analysis
- **Software and Programming Languages**: MATLAB & Simulink, ROS 2, ROS, MuJoCo, SolidWorks, Autodesk AutoCAD, Autodesk Fusion 360, Arduino IDE, Python, C, C++, LaTeX, VESC tool, Autodesk Eagle

Key Courses Taken

- **Robotics, Systems and Control**: Optimal Control, Modern Control Theory, Probabilistic Artificial Intelligence, Integer Optimization, Image Analysis and Computer Vision, Robot Dynamics, Nonlinear Dynamics and Chaos I, Computational Models of Motion, Multivariate Data Analysis for Process Modeling, Signal Processing, Parallel Scientific Computing.
- **Design & Mechanics**: Kinematics and Dynamics of Machinery, Mechanics of Human Movement, Theory of Vibrations, Fundamentals of Finite Element Analysis, Optimization Methods in Mechanical Design.